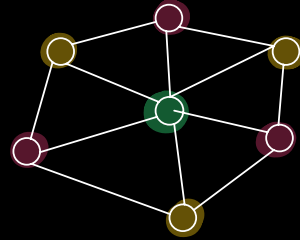
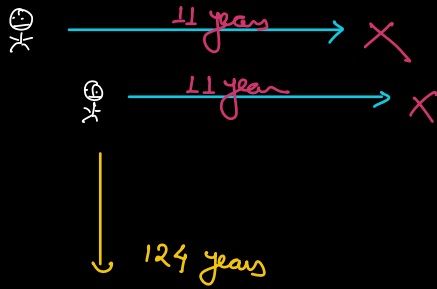
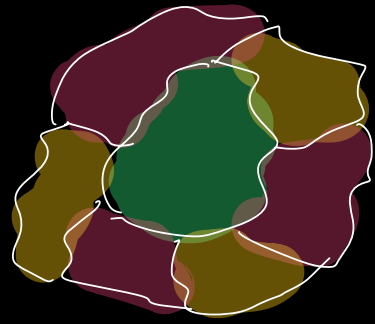
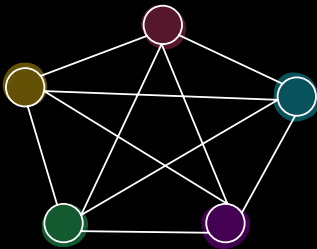


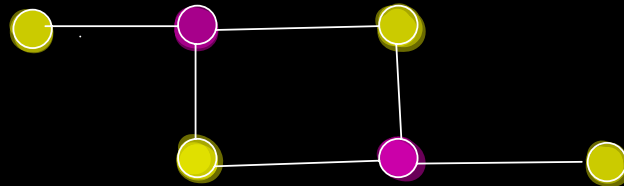
De-Morgan
Set Theory
Math. Induction



First computer assisted proof of anything in any domain.

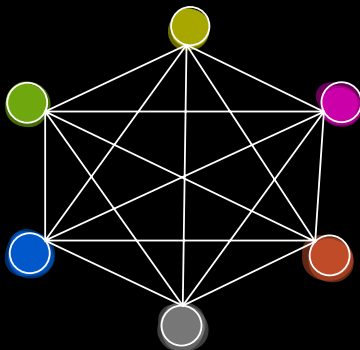


5



2, 3, 4, 5, 6 —

Chromatic No. - Min no of colors required to color a graph.



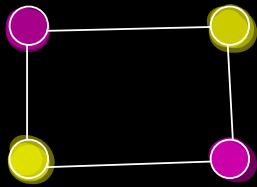
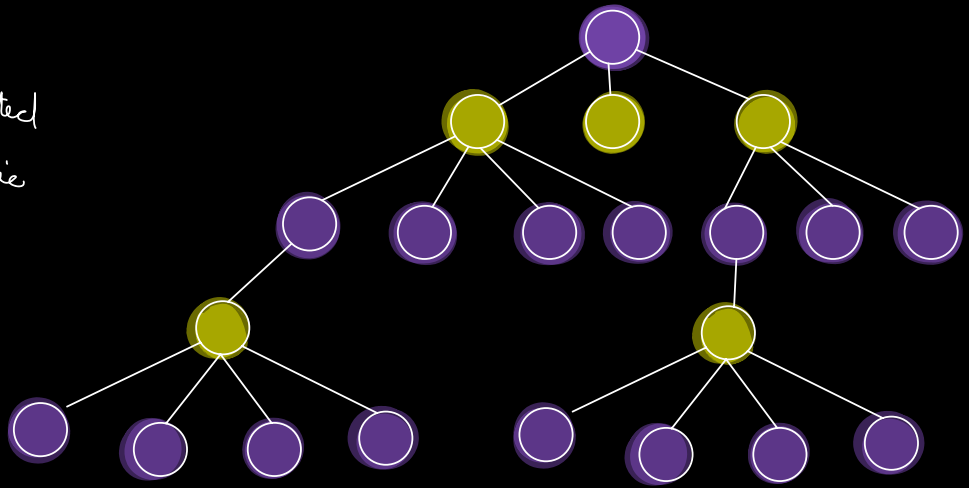
6

Q Find the Chromatic no of a graph. $\equiv$

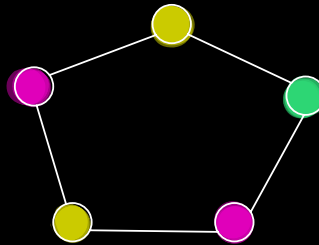
Tree

- Connected
- Acyclic

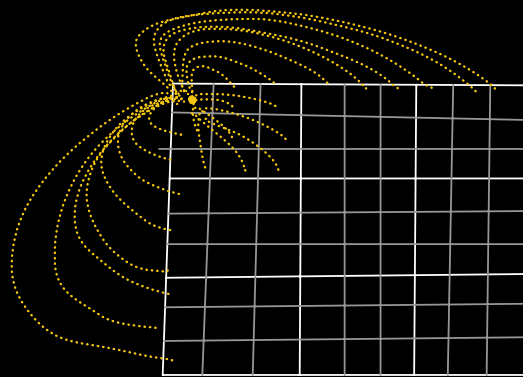
Chromatic No : 2



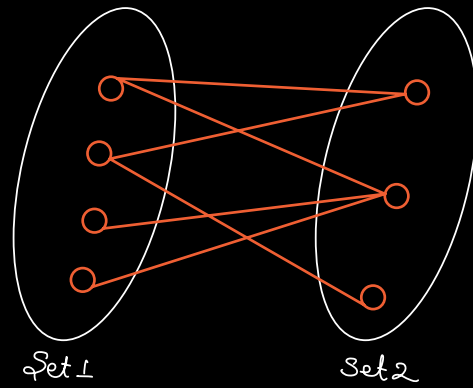
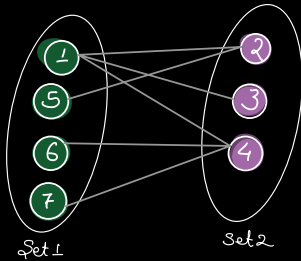
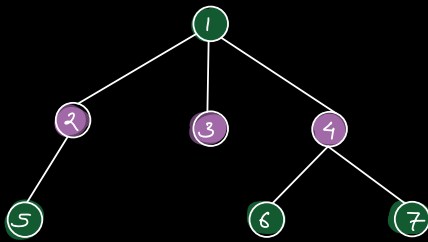
$|V| \rightarrow \text{even}$
Ch.No $\rightarrow 2$



$|V| \rightarrow \text{odd}$
Ch.No. $\rightarrow 3$

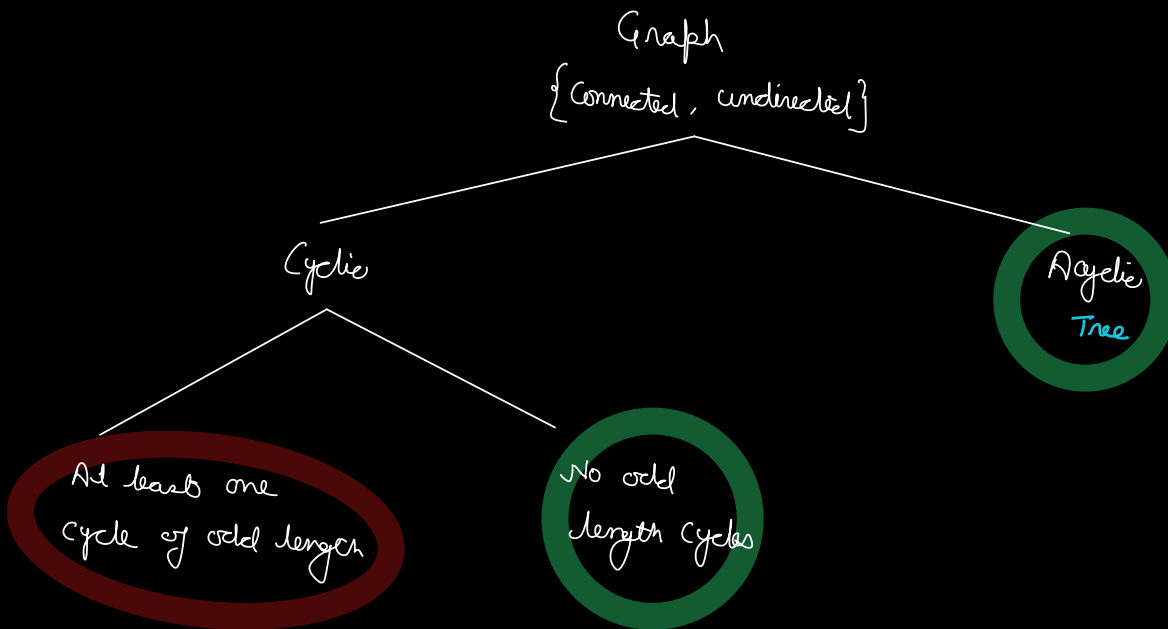


Bipartite Graphs

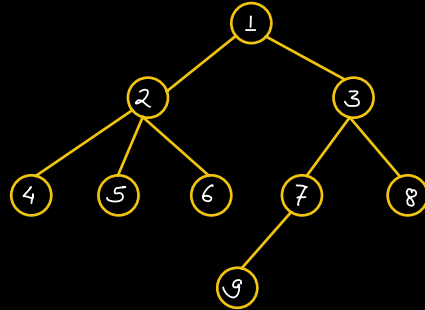


Chromatic No of
Bi-partite : 2

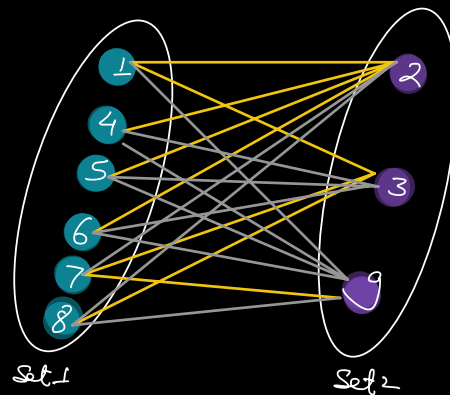
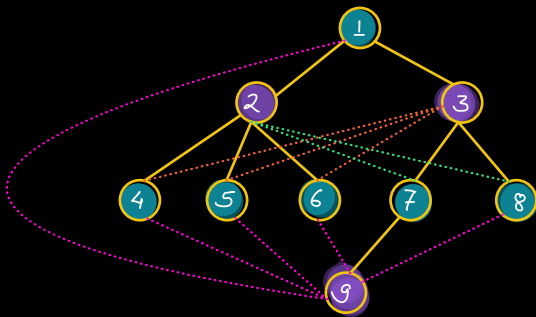
Q Check if a graph is bi-partite?



Google Q Given a tree. Return the max no. of edges that can be added in the tree such that the resulting graph is bi-partite.



Ans: 10



$$\text{Total possible edges} = |\text{Set1}| \times |\text{Set2}|$$

$$\text{No. of already existing edges} = (N-1)$$

$$\text{No. of edges that can be added} = |\text{Set1}| \times |\text{Set2}| - (N-1)$$

Uber Q

Extroverts doing party

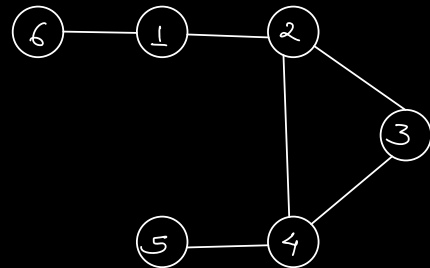
Only the people who have at least K friends can join the party (only if those K friends are also joining)

Person

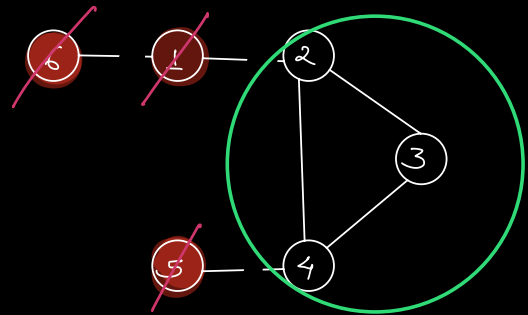
Friends

1	:	{2, 6}
2	:	{1, 3, 4}
3	:	{2, 4}
4	:	{3, 5, 2}
5	:	{4}
6	:	{1}

$K = 2$



1	:	{2, 6}	
2	:	{ 1 , 3, 4}	✓
3	:	{2, 4}	✓
4	:	{3, 5 , 2}	✓
5	:	{4}	
6	:	{1}	



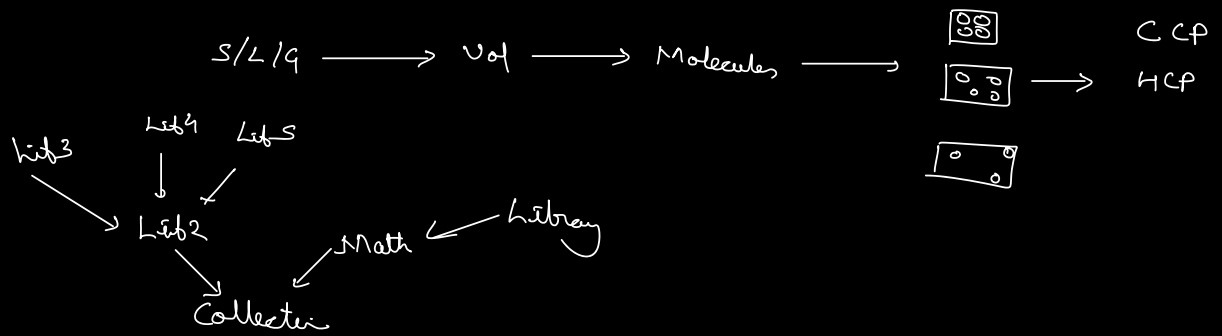
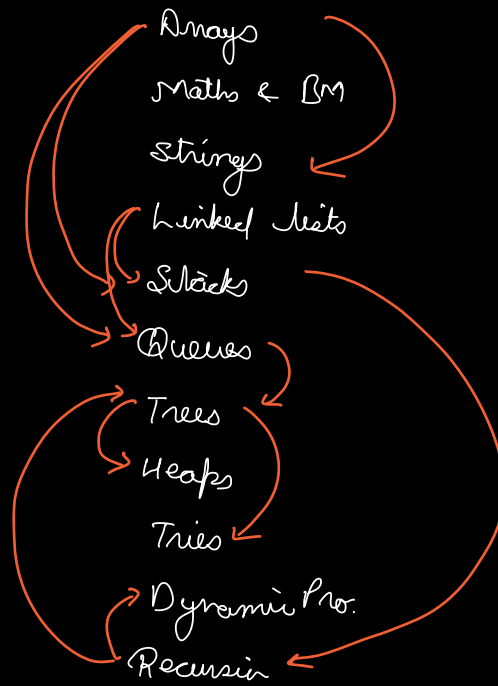
People	1	2	3	4	5	6
# friends	2	3	2	3	1	1



Multisourced BFS

TC : $O(V+E)$

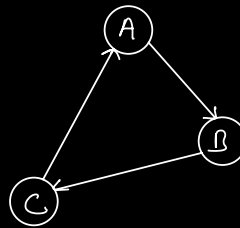
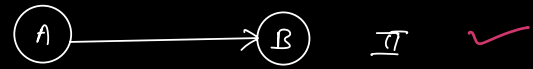
Graphs
BFS
DFS
Cycle detection
Coloring
Bi-Partite



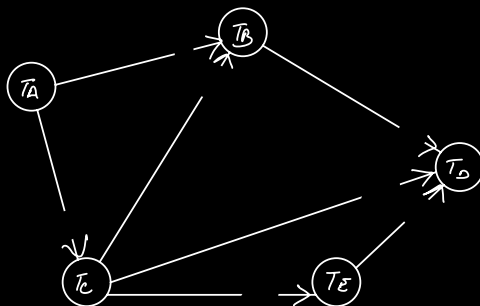
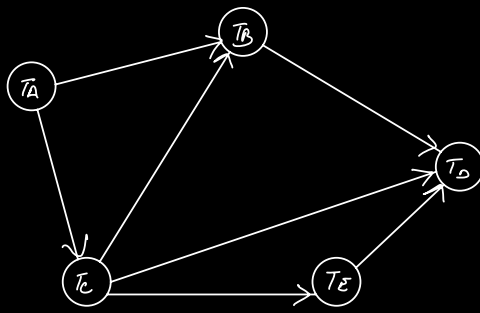
Maven
NPM
Gradle

Topological Sorting

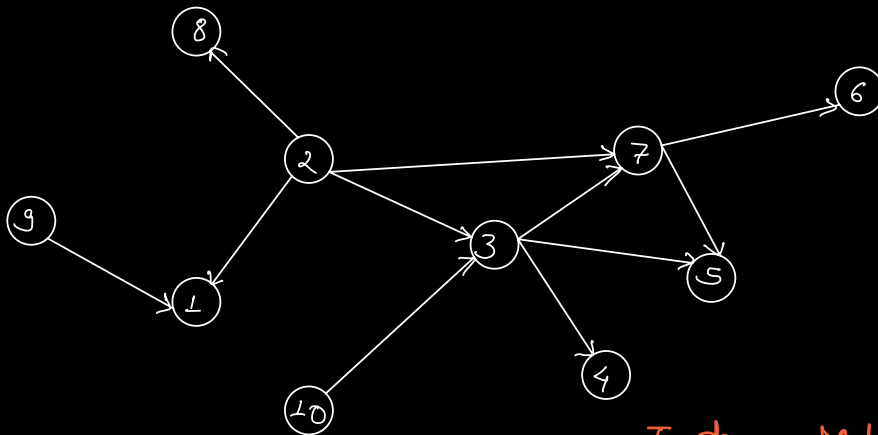
- Directed
- Acyclic



Dead lock



$T_A \rightarrow T_C \rightarrow T_E \rightarrow T_B \rightarrow T_D$



Adj. list (outgoing edges)

1 : { }
 2 : { 1, 3, 7, 8 }
 3 : { 4, 5, 7 }
 4 : { }
 5 : { }
 6 : { }
 7 : { 6, 5 }
 8 : { }
 9 : { 1 }
 10 : { 3 }

In degree Map
Vertex : Indegree

1 : ~~2~~ 0
 2 : 0
 3 : ~~2~~ 0
 4 : ~~1~~ 0
 5 : ~~2~~ 1 0
 6 : ~~1~~ 0
 7 : ~~2~~ ~~1~~ 0
 8 : ~~1~~ 0
 9 : 0
 10 : 0

> 97%

11:20pm

~~2, 9, 10, 8, 1, 3, 4, 7, 6, 5~~

Order : 2, 9, 10, 8, 1, 3, 4, 7, 6, 5